



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to RG FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

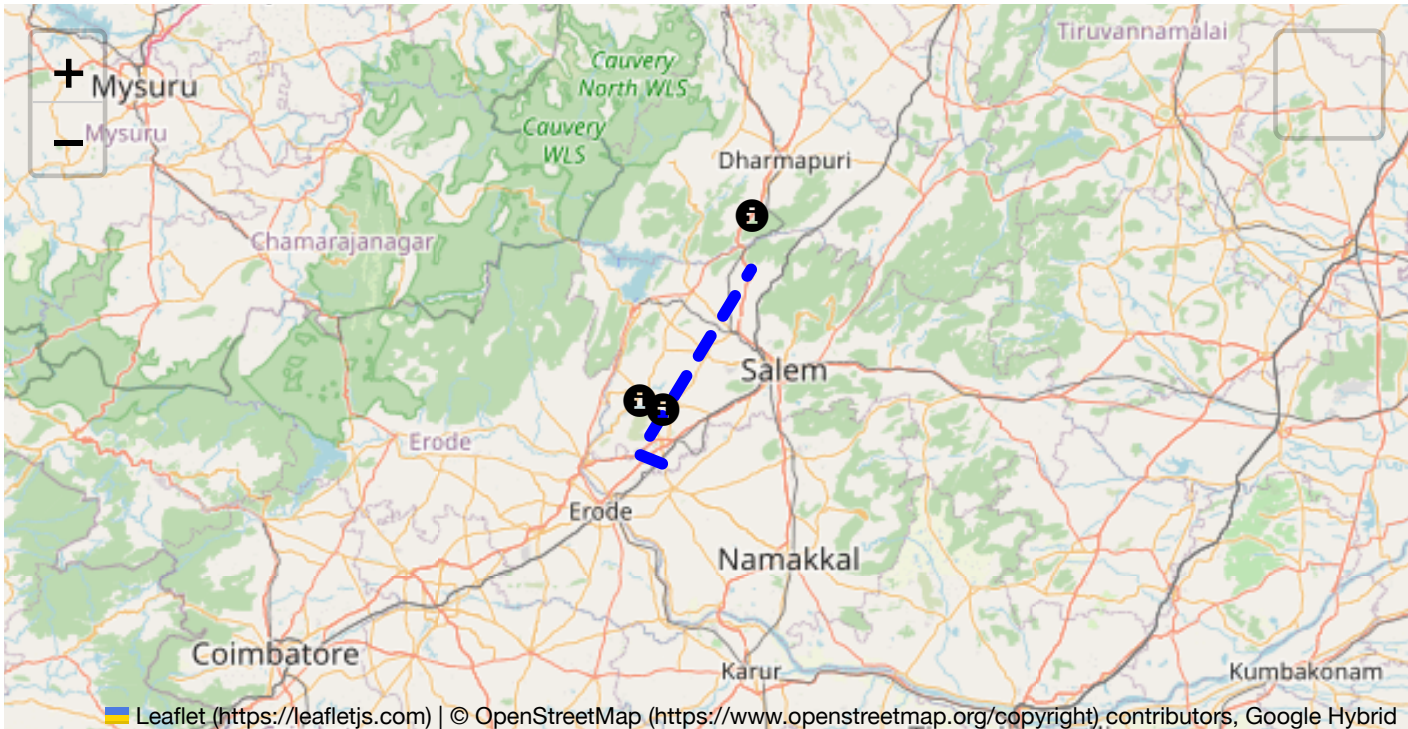
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

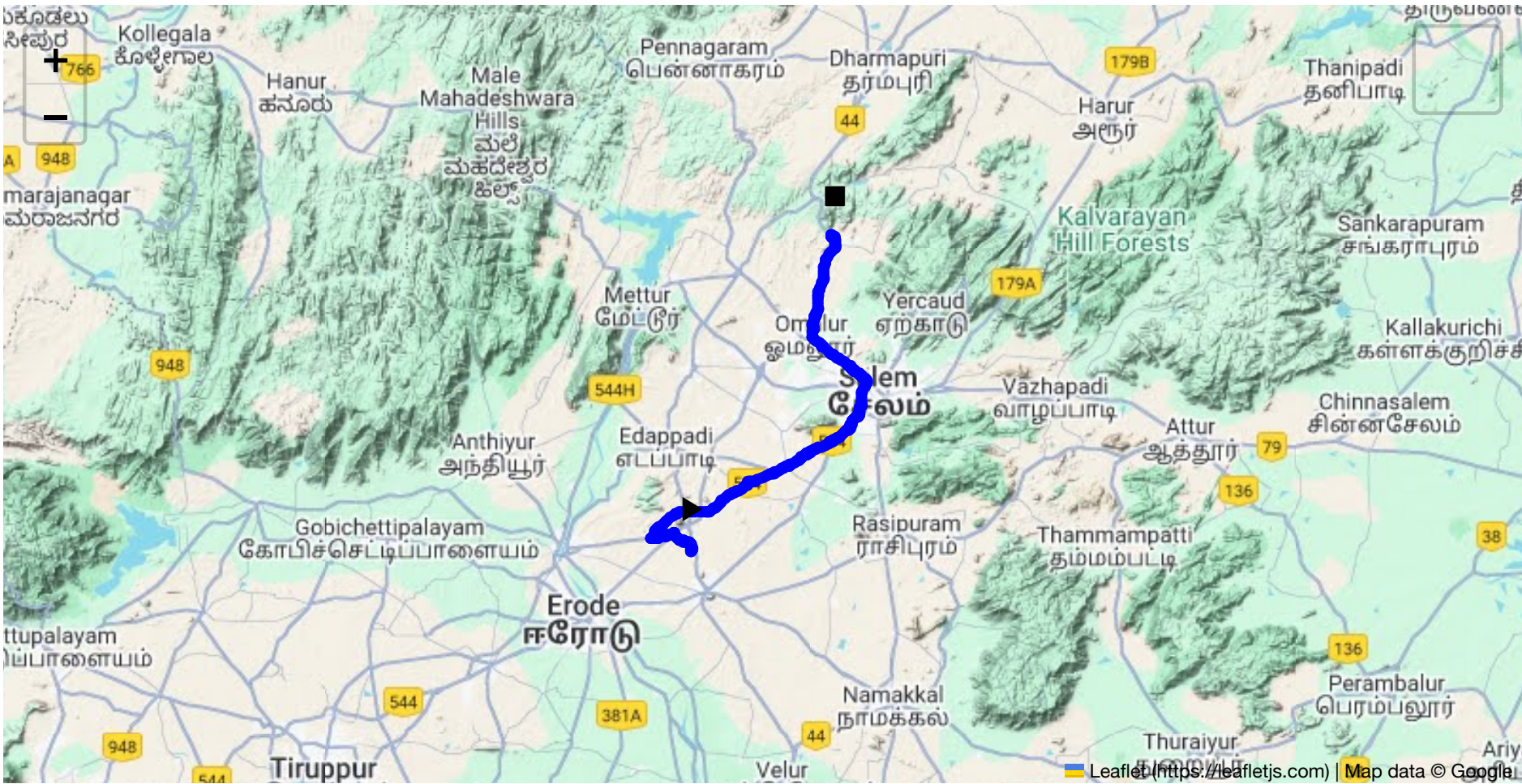
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 83.02 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours

Start: (11.4381, 77.8734)
End: (11.882019, 78.081192)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, to V3JJ+RF Kadayampatti via FR48+7XQ, Bhavani Main Road, covers approximately 83.02 kilometers. The journey is typically straightforward, moving through rural towns and along state highways. Bhavani Main Road is an essential connection in this route, providing a direct path with intersections of rural and urban roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate, with significant rainfall during the monsoon season (June to September) and occasional cyclones. During these periods, heavy rains can cause road flooding and reduce visibility. The dry season, from October to May, might see high temperatures, which can lead to road surface deformities like softening asphalt.

3. Analysis of Traffic Patterns

The route experiences moderate to heavy traffic, particularly in the morning (7-9 AM) and evening (5-8 PM) due to local commuters. The convergence at Bhavani Main Road can be particularly congested. Congestion-prone areas also include intersections with smaller rural roads and town centers.

4. Assessment of Road Quality and Infrastructure

Overall, the road quality ranges from fair to good. However, certain sections, particularly those moving through smaller towns, may have narrower lanes with inconsistent maintenance. Potholes and uneven surfaces are common in less urbanized stretches. Signage varies, with better quality in urbanized zones.

5. Suggestions for Alternative Routes for Emergencies

In case of road closure or severe congestion, an alternative can be taking the NH44 after joining from a suitable feeder road from Padaiveedu or nearby. This highway is generally well maintained and can provide a reliable detour, albeit with increased travel time.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu requires permits for hazardous material transport, with restrictions on movement through heavily populated areas during the day to reduce accident risks. Proper signage on vehicles and adherence to designated routes are mandatory.

7. Overview of Historical Incidents

While the specific route has limited documented incidents, areas around industrial hubs on Bhavani Main Road have reported incidents involving chemical spills from heavy vehicles. Ensuring driver awareness and equipment checks could mitigate such risks.

8. Environmental Considerations and Sensitive Areas

The route crosses regions with agricultural activities. Drivers must be cautious not to contaminate the environment, especially near water bodies and fields. Awareness of local wildlife crossings, particularly in less urban areas, is crucial for environmental conservation.

9. Analysis of Communication Coverage

The communication coverage along the route is generally reliable in urban areas; however, some rural stretches suffer from reduced connectivity or dead zones. It would be advisable to check mobile network maps for specific providers prior to travel.

10. Estimated Emergency Response Times

Emergency response in urban areas can be prompt (approximately 15-20 minutes), while response times in rural stretches may extend to 30-45 minutes due to distance from facilities and potential route obstructions.

11. Overall Summary of Risk Assessment

The route, while feasible for transporting hazardous materials, presents certain risks due to weather, road quality, and congestion. With adherence to local regulations and preparedness for environmental hazards, risks can be mitigated for safe transit. Regular updates regarding traffic and weather conditions, as well as proper communication tools and emergency protocols, will support a secure journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.8848607, 78.07615539999999	10 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45358, 77.85781	30 KM/Hr
6	Turn	High	11.45353, 77.85697	15 KM/Hr
7	Turn	Medium	11.45853, 77.85139	30 KM/Hr
8	Turn	Medium	11.45906, 77.85132	30 KM/Hr
9	Turn	Medium	11.46307, 77.84941	30 KM/Hr
10	Turn	Medium	11.46316, 77.84921	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
13	Turn	Medium	11.64348, 78.12002	30 KM/Hr
14	Turn	High	11.69388, 78.10965	15 KM/Hr
15	Turn	High	11.69400, 78.10963	15 KM/Hr
16	Turn	High	11.88486, 78.07616	15 KM/Hr
17	Blind Spot	Blind Spot	11.88494, 78.07624	10 KM/Hr
18	Turn	Medium	11.88197, 78.07989	30 KM/Hr
19	Turn	Medium	11.88202, 78.08088	30 KM/Hr
20	Turn	High	11.88220, 78.08108	15 KM/Hr
21	Turn	High	11.88217, 78.08122	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
15	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
16	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
18	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
19	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
20	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
21	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
22	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
23	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
26	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
28	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
29	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
31	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
17	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
24	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
25	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
27	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
30	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
32	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.8848607, 78.07615539999999



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



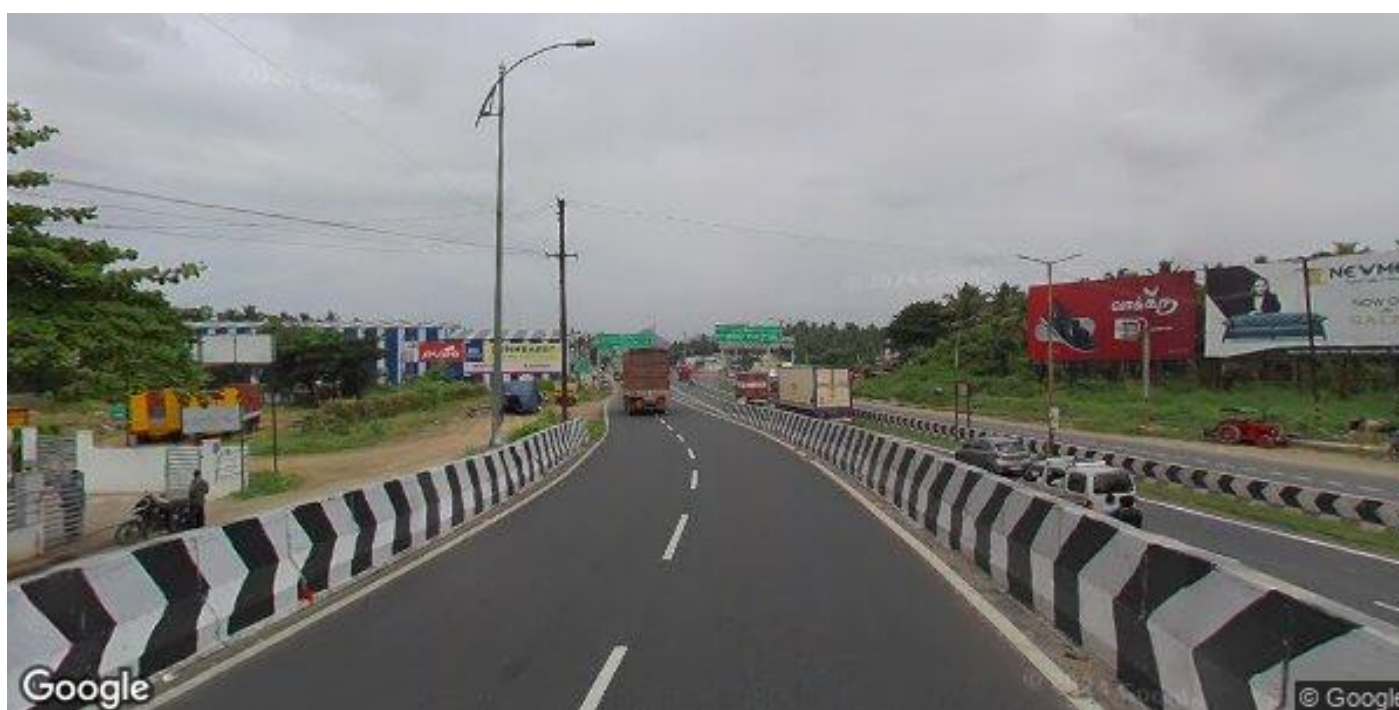
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64348, 78.12002



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69388, 78.10965



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69400, 78.10963



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.88486, 78.07616



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.88494, 78.07624



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88197, 78.07989
